

Expectation and Variance in the General Setting

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L_p 空间就是“那些你把 $|X|^p$ 积分后能得到有限值的随机变量的集合”。

$L_p(\Omega, \mathcal{A}, P)$ = “不会太危险”的一类随机变量。

1 Expectation and Variance

Definition 1.1 (L^p -space) Let $(\Omega, \mathcal{A}, P)$ be a probability space. For $1 \leq p < \infty$ we define

$$L^p(\Omega, \mathcal{A}, P) := \{X : \Omega \rightarrow \mathbb{R} \mid X \text{ measurable and } \int_{\Omega} |X|^p dP < \infty\}.$$

For $p = \infty$ we write L^∞ for the space of essentially bounded random variables.

Definition 1.2 (Expectation & Moments) If $X \in L^1(\Omega, \mathcal{A}, P)$, its expectation (or first moment) is

$$\mathbb{E}[X] := \int_{\Omega} X dP = \int_{\mathbb{R}} x dP_X(x).$$

More generally, if $k \in \mathbb{N}$ and $X^k \in L^1$, the k -th moment is

所以这个一般是计算期望的前提，避免期望爆炸

$$\mathbb{E}[X^k] = \int_{\Omega} X^k dP.$$

Definition 1.3 (Variance & Covariance) For $X, Y \in L^2(\Omega, \mathcal{A}, P)$ the variance and covariance are defined by

L_2 是计算Var和Cov的前提

$$\text{Var}(X) := \mathbb{E}[(X - \mathbb{E}[X])^2], \quad \text{Cov}(X, Y) := \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])].$$

2 Markov and Chebyshev Inequalities

2.1 Cauchy–Schwarz Inequality

Theorem 2.1 (Cauchy–Schwarz Inequality) Let $x, y \in L^2(\Omega, \mathcal{A}, P)$. Then,

$$|\mathbb{E}[xy]|^2 \leq \mathbb{E}[x^2] \mathbb{E}[y^2].$$

2.2 Markov Inequality

Theorem 2.2 (Markov Inequality) Let $g : [0, \infty) \rightarrow [0, \infty)$ be a non-decreasing measurable function and let X be a non-negative r.v. Then for every $a > 0$

$$P\{X \geq a\} \leq \frac{\mathbb{E}[g(X)]}{g(a)}.$$

In particular, with $g(x) = x$ we obtain $P\{X \geq a\} \leq \frac{\mathbb{E}[X]}{a}$.

2.3 Chebyshev Inequality

Theorem 2.3 (Chebyshev) For any $X \in L^2$ and $\varepsilon > 0$

$$P\{|X - \mathbb{E}[X]| \geq \varepsilon\} \leq \frac{\text{Var}(X)}{\varepsilon^2}.$$

Chebyshev's inequality provides a distribution-free upper bound on the probability of large deviations. It is a key tool in proving the Weak Law of Large Numbers.

3 Probability Distributions

3.1 Discrete Distributions

Definition 3.1 (Uniform Distributions on $\{1, \dots, n\}$) A discrete r.v. X is uniform on $\{1, \dots, n\}$ if $P\{X = i\} = \frac{1}{n}$ for each i .

Definition 3.2 (Binomial Distributions $\text{Bin}(n, p)$) Let $n \in \mathbb{N}$ and $p \in (0, 1)$. If X counts the number of heads in n independent Bernoulli(p) trials then

$$P\{X = k\} = \binom{n}{k} p^k (1-p)^{n-k}, \quad k = 0, \dots, n.$$

Definition 3.3 (Poisson Distributions $\text{Pois}(\lambda)$) For $\lambda > 0$, a r.v. X is Poisson with rate λ if

Parameter $\lambda > 0$

$$P\{X = k\} = e^{-\lambda} \frac{\lambda^k}{k!}, \quad k \in \mathbb{N}_0.$$

It often models the number of arrivals in a fixed time interval.

3.2 Continuous Distributions

Definition 3.4 (Uniform on $[a, b]$)

A continuous r.v. X is uniform on $[a, b]$ if its density is

$$f_X(x) = \begin{cases} (b-a)^{-1}, & x \in [a, b], \\ 0, & \text{otherwise.} \end{cases}$$

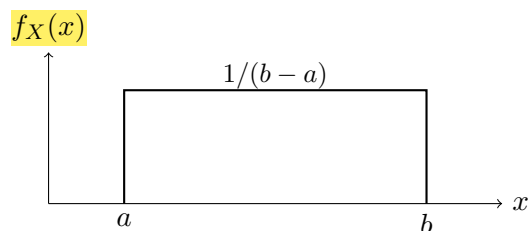


Figure 1: Density of a continuous uniform distribution on $[a, b]$.

Definition 3.5 (Normal $\mathcal{N}(\mu, \sigma^2)$) A r.v. X is normal with mean μ and variance $\sigma^2 > 0$ if its density is

$$f_{\mu, \sigma}(x) = \frac{1}{\sqrt{2\pi} \sigma} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right).$$

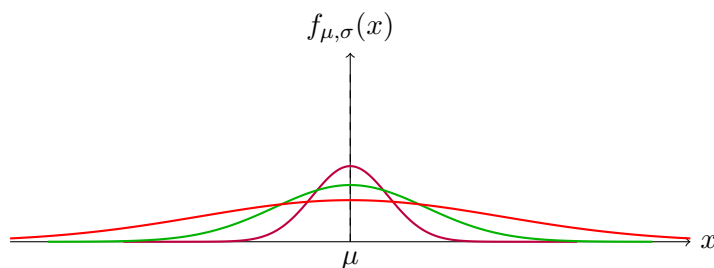


Figure 2: Normal densities with identical mean μ and different variances ($\sigma_{\text{orange}} < \sigma_{\text{green}} < \sigma_{\text{red}}$).

4 Multivariate Normal Distribution

Let $X = (X_1, \dots, X_n)^\top \in \mathbb{R}^n$ with mean vector μ and covariance matrix Σ . We write $X \sim \mathcal{N}(\mu, \Sigma)$ if

$$f_{\mu, \Sigma}(x) = \frac{1}{\sqrt{(2\pi)^n \det \Sigma}} \exp\left(-\frac{1}{2}(x - \mu)^\top \Sigma^{-1}(x - \mu)\right).$$

Key facts.

- Σ is symmetric positive semi-definite and thus possesses an eigen-decomposition $\Sigma = Q\Lambda Q^\top$.
- The contour ellipsoids of $f_{\mu, \Sigma}$ are aligned with the eigenvectors of Σ .
- Independence of components X_i is equivalent to Σ being diagonal.
- If $X \sim \mathcal{N}(\mu_1, \Sigma_1)$ and $Y \sim \mathcal{N}(\mu_2, \Sigma_2)$ are independent, then $X + Y \sim \mathcal{N}(\mu_1 + \mu_2, \Sigma_1 + \Sigma_2)$.

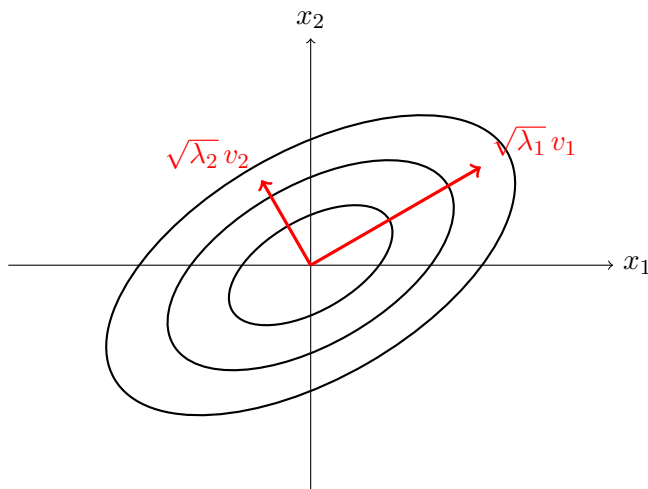


Figure 3: Contours of a bivariate normal distribution with mean $(0, 0)$. Ellipses represent constant Mahalanobis distance. Red arrows indicate eigenvectors v_1, v_2 of Σ scaled by $\sqrt{\lambda_1}$ and $\sqrt{\lambda_2}$.

5 Mixture of Gaussians

Definition 5.1 (Gaussian Mixture Model) Let $\{\pi_i\}_{i=1}^k$ be non-negative weights satisfying $\sum_{i=1}^k \pi_i = 1$, and let f_{μ_i, Σ_i} denote Gaussian densities. The Gaussian mixture density is

$$f(x) = \sum_{i=1}^k \pi_i f_{\mu_i, \Sigma_i}(x).$$

GMMs combine multiple Gaussian “clusters” and can approximate arbitrary continuous densities. The Expectation–Maximisation (EM) algorithm is the canonical method for parameter estimation.

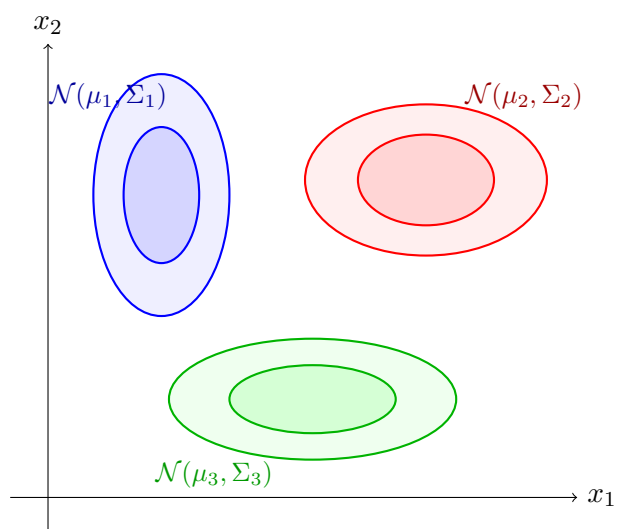


Figure 4: Contour plot of a 3-component Gaussian mixture in $\mathbb{R}^2$. Shaded ellipses depict 1σ and 2σ level sets for each component.

6 Additional Results

6.1 Weak Law of Large Numbers

Theorem 6.1 (WLLN) *Let $X_1, X_2, \dots$ be i.i.d. with $\mathbb{E}[X_i] = \mu$ and $\text{Var}(X_i) = \sigma^2 < \infty$. Then for any $\varepsilon > 0$*

$$\Pr\left\{\left|\frac{1}{n}\sum_{i=1}^n X_i - \mu\right| > \varepsilon\right\} \longrightarrow 0 \quad (n \rightarrow \infty).$$

6.2 Central Limit Theorem

Theorem 6.2 (CLT) *Under the same assumptions as the WLLN,*

$$\frac{\sqrt{n}}{\sigma}\left(\frac{1}{n}\sum_{i=1}^n X_i - \mu\right) \xrightarrow{d} \mathcal{N}(0, 1).$$